



Algorithms as work designers: How algorithmic management influences the design of jobs

Xavier Parent-Rochelleau ^a  , Sharon K. Parker ^b

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Highlights

- The article describes six management functions that AI-enabled algorithms are able to perform.
- The influence of each AM function on work design characteristics is reviewed.
- We review and discuss parameters likely to mitigate the effects of AM.

Abstract

We review the literature on algorithmic management (AM) to bridge the gap between this emerging research area and the well-established theory and research on work design. First, we identify six management functions that algorithms are currently able to perform: monitoring, goal setting, performance management, scheduling, compensation, and job termination. Second, we show how each AM function affects key job resources (e.g., job autonomy, job complexity) and key job demands (e.g., workload, physical demands); with each of these resources and demands being important drivers of worker motivation and their well-being. Third, rejecting a deterministic perspective and drawing on sociotechnical systems theory, we outline key categories of variables that moderate the link between AM on work design, namely transparency, fairness and human

influence (e.g., whether workers can control the system). We summarize our review in the form of a model to help guide research on AM, and to support practitioners and designers in the creation and maintenance of meaningful jobs in the era of algorithms.

Introduction

Recent advances in artificial intelligence technologies have a considerable influence on people's work. Experts argue that the digitalization of workplaces is unprecedented and that we are in the “fourth industrial revolution” (de Ruyter, Brown, & Burgess, 2019; Ghislieri, Molino, & Cortese, 2018; Schwab, 2017). Researchers have begun to focus on the repercussions of these technological advances for HRM and management (Kellogg, Valentine, & Christin, 2019; Lindebaum, Vesa, & den Hond, 2019; Minbaeva, 2020; Murray, Rhymer, & Sirmon, 2020; Raisch & Krakowski, 2021; Tambe, Cappelli, & Yakubovich, 2019). Since most of the existing jobs are likely to be heavily transformed in the following years (Jarrahi, 2018; Raisch & Krakowski, 2021; Roos & Shroff, 2017), and many new forms of jobs are expected to appear (Paus, 2018; West, 2018), it is vital to ensure that the design of digital work promotes human well-being (Parker & Grote, 2020; Schroeder, Bricka, & Whitaker, 2019).

Fortunately, to help guide the design of digitalized work, there is a great deal of well-established theory and evidence regarding the work design parameters that lead to meaningful jobs (Hackman & Oldham, 1976; Humphrey, Nahrgang, & Morgeson, 2007; Parker, Morgeson, & Johns, 2017). Work design refers to “the content and organization of one's work tasks, activities, relationships, and responsibilities” (Parker, 2014, p. 47). This body of research has provided extensive evidence for the influence of work design characteristics such as job autonomy and task variety on employee's motivation and well-being, as well as on key HR indicators such as turnover and productivity (Chang, Wang, & Huang, 2013; Humphrey et al., 2007; Parker, 2014; Parker & Turner, 2002; Torraco, 2005). Scholars (e.g., Parker and Grote (2020); Demerouti (2020) have recently sought to apply work design research and theory to understand the impact of the digital revolution on people's work. We build on this perspective here, focusing specifically on understanding the impact of algorithmic management on work design and, as a consequence, on important outcomes such as well-being and motivation.

The management of employees via algorithms is one of the most disruptive forms of technological change currently being implemented (Duggan, Sherman, Carbery, & McDonnell, 2020; Kellogg et al., 2019; Lee, 2018; Rosenblat, 2018; Wesche & Sonderegger, 2019). Algorithmic management refers to “a system of control where [...] algorithms are given the responsibility for making and executing decisions affecting labor, thereby limiting human involvement and oversight of the labor process”

(Duggan, Sherman, Carbery, & McDonnell, 2020, p. 6).

Existing research on algorithmic management tends to suggest that it generates more negative than positive outcomes for workers. In particular, AM has been associated with a reduction of workers' autonomy (Iahagi, Croteau, Audebrand, & Marsan, 2019; Mohlmann & Zalmanson, 2017; Toyoda

autonomy (Jabagi, Croteau, Audubert, & Marsali, 2019; Mohammad & Zammaran, 2017; Royoua, Lucas, & Gratch, 2020), and the creation of power asymmetry in information availability (Calo & Rosenblat, 2017; Jarrahi & Sutherland, 2018; Rosenblat & Stark, 2016; Shapiro, 2018). These consequences potentially explain why AM has also been shown to result in negative emotions, unfairness perceptions, low trust (Lee, 2018; Zarsky, 2016), psychological contract breach (Tomprou & Lee, 2019), low job satisfaction (Brawley & Pury, 2016; Griesbach, Reich, Elliott-Negri, & Milkman, 2019), lower life satisfaction (Keith, Harms, & Tay, 2019), and reduced engagement in work (Bucher, Fieseler, & Lutz, 2019). This apparently dark portrait of AM, however, warrants additional research, and has limitations. First, the emerging literature on the topic is spread across different streams of research and conceptual approaches, and has mainly examined one algorithmic function at the time, lacking theoretical integration. Second, existing research often seems to be based on deterministic assumptions about the consequences of technology, implying pre-determined outcomes. Third, empirical research has focused on the context of gig work, such that AM in traditional work contexts has been overlooked.

In this paper, we address these limitations by reviewing the findings from empirical and descriptive studies of AM practices, drawing on work design theory to organize our analysis. We propose a holistic framework for understanding the effects of AM on work design, including moderators of these effects, and the associated outcomes. This framework can be tested and expanded to support systematic and theory-driven future research on the topic of AM (see Fig. 1). Specifically, we first identify six HR and management functions that organizations can automate with algorithms (i.e. monitoring, goal setting, performance management, scheduling, compensation, and job termination). Second, drawing on existing evidence, we propose how these AM functions affect work design, in terms of key job resources and job demands. As indicated in our model, considerable evidence shows that these job resources and demands, in turn, affect workers' motivation, well-being, and performance (we do not review this evidence, but refer the reader to existing meta-analyses and reviews). Third, adopting a voluntarist rather than deterministic standpoint, we propose several moderators of this relationship between AM and work design (and hence outcomes). To identify these moderators, we draw on the socio-technical systems perspective of work design (Emery & Trist, 1978; Makarius, Mukherjee, Fox, & Fox, 2020). Altogether, the proposed model can not only can guide future research, but also can be used to help HR managers and system designers to ensure that the procurement and implementation of algorithmic management is done in a way that maintains or creates motivating work for workers.

In what follows, we describe algorithms AND the functions of management they can perform, followed by how they affect work design. We then turn our attention to what we label as

'sociotechnical moderators', or the most promising parameters that can shape the influence of AM on work design, thereby providing a path for creating well-designed jobs.

Section snippets

Algorithms AND the functions of management they perform

An algorithm is a “computational formula that autonomously makes decisions based on statistical models or decision rules without explicit human intervention, [such as] a sequence of instructions telling a computer what to do within a set of precisely-defined steps and rules designed to accomplish a task” (Duggan, Sherman, Carbery, & McDonnell, 2020, p. 6). Algorithms are used in many data-based systems used in HRM and management (Cheng & Hackett, 2019; Duggan, Sherman, Carbery, & McDonnell, 2020 ...

The effect of algorithmic management on work design

Among the different conceptualizations of work design (see Parker et al., 2017 for a review), we draw mainly on the popular job demands-resources model (J D-R model: Demerouti, Bakker, Nachreiner, & Schaufeli, 2001), emphasizing the importance of considering both job resources and job demands when designing work (Demerouti et al., 2001). Considerable evidence links job resources and job demands with workers' motivation, well-being and performance (for meta-analyses or reviews, see: Humphrey et ...

How can organizations ensure that am enables and sustains well-design jobs?

The above discussion of AM functions and their consequences suggests that the outcomes for work design are principally negative, with AM tending to lower job resources and to increase job demands. However, we assert that focusing solely on the negative consequences of algorithmic management is not a fruitful long-term approach. Rather, we embrace a (moderately) voluntarist rather than deterministic approach to apprehend the consequences of algorithmic management. As summarized by Strohmeier ...

Discussion and conclusion

The contributions of our model and review are multiple. First, albeit preliminary, the evidence reviewed clearly highlights the influence of AM on work design characteristics, with most of this influence appearing to be negative. In unpacking the phenomenon of algorithmic management and its six (actual) functions, we develop a more holistic approach of algorithmic management, allowing its examination across the distinctly operating components rather than as a monolithic piece. Several ...

Conclusion

Building on previous research on algorithmic management, and drawing on well-established conceptual frameworks (job demands-resources model and sociotechnical systems design), we have proposed a conceptual model to enhance knowledge development and application about AM. We argue that such model is necessary at this stage of the development of the discipline in order to guide research that can leverage concrete and tangible improvements in AM implementation in organizations. ...

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